


PHYSICS
STANDARD LEVEL
PAPER 2
Topic A.2.3 Circular Motion and D.1 Gravitation Study Guide

May 2003 - 2013

1 hour 15 minutes

Candidate session number

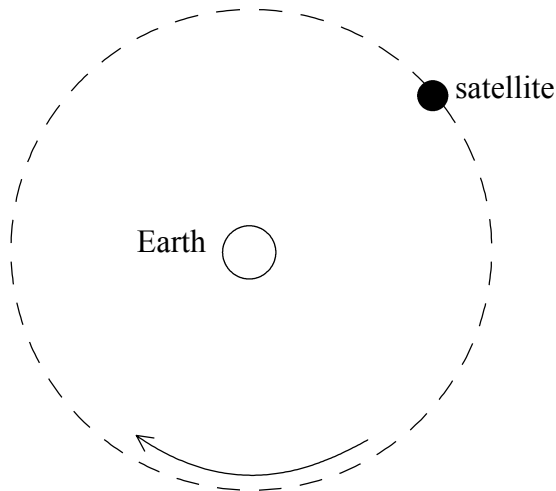
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INSTRUCTIONS TO CANDIDATES

- Write your session number in the boxes above.
- Do not open this examination paper until instructed to do so.
- Section A: answer all of Section A in the spaces provided.
- Section B: answer one question from Section B in the spaces provided.
- At the end of the examination, indicate the numbers of the questions answered in the candidate box on your cover sheet.



A2. A satellite orbits the Earth at constant speed as shown below.



(a) Draw on the diagram

(i) an arrow labelled F to show the direction of the gravitational force of the Earth on the satellite.

(ii) an arrow labelled V to show the direction of the velocity of the satellite. [2]

(b) Although the speed of the satellite is constant, it is accelerating. Explain why it is accelerating. [2]

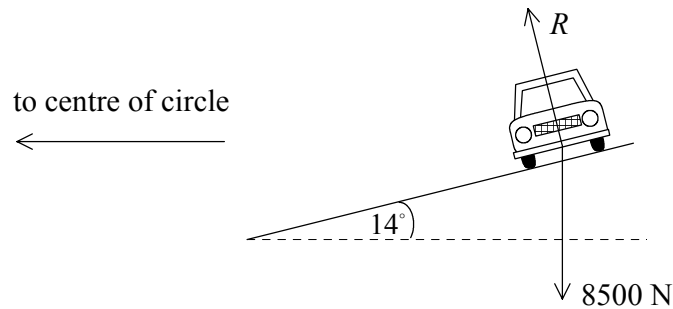
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(c) Discuss whether or not the gravitational force does work on the satellite. [3]

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A2. This question is about the motion of a car.

A car of weight 8500 N is travelling at constant speed along a road that is an arc of a circle. In order that the car may travel more easily round the arc, the road is banked at 14° to the horizontal, as shown below.



At one particular speed v of the car, there is no frictional force at 90° to the direction of travel of the car between the tyres and the road surface. The reaction force of the road on the car is R .

- (a) Deduce that the horizontal component of the force R is approximately 2100 N. [2]

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- (b) State the magnitude and direction of the resultant force acting on the car. [2]

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- (c) Determine the speed v of the car at which it travels round the arc of radius 150 m without tending to slide. [3]

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- (d) Deduce in which direction the car will tend to slide if it travels round the curve at a speed greater than v . [2]

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SECTION B

*This section consists of three questions: B1, B2 and B3. Answer **one** question.*

- B1.** This question is in **two** parts. **Part 1** is about the kinematics and dynamics of circular motion. **Part 2** is about the electric field due to a charged sphere and the motion of electrons in that field.

Part 1 Circular motion

- (a) A car goes round a curve in a road at constant speed. Explain why, although its speed is constant, it is accelerating. [2]

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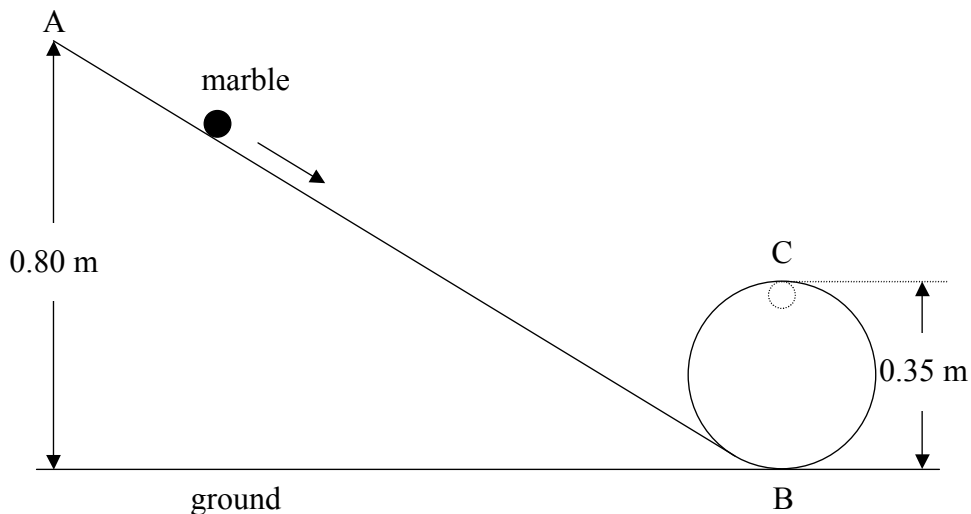
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In the diagram below, a marble (small glass sphere) rolls down a track, the bottom part of which has been bent into a loop. The end A of the track, from which the marble is released, is at a height of 0.80 m above the ground. Point B is the lowest point and point C the highest point of the loop. The diameter of the loop is 0.35 m.



The mass of the marble is 0.050 kg. Friction forces and any gain in kinetic energy due to the rotating of the marble can be ignored. The acceleration due to gravity, $g = 10 \text{ ms}^{-2}$.

Consider the marble when it is at point C.

(This question continues on the following page)

(Question B1, part 1 continued)

(b) (i) On the diagram opposite, draw an arrow to show the direction of the resultant force acting on the marble. [1]

(ii) State the names of the **two** forces acting on the marble. [2]

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(iii) Deduce that the speed of the marble is 3.0 ms^{-1} . [3]

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(iv) Determine the resultant force acting on the marble and hence determine the reaction force of the track on the marble. [4]

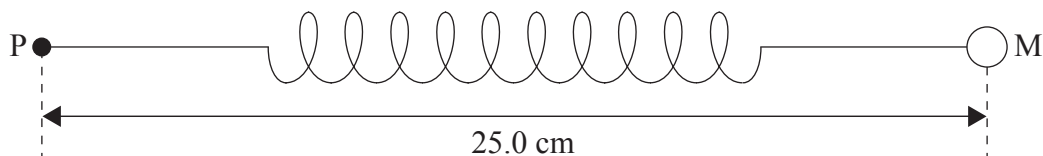
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A2. This question is about circular motion.

A linear spring of negligible mass requires a force of 18.0 N to cause its length to increase by 1.0 cm.

A sphere of mass 75.0 g is attached to one end of the spring. The distance between the centre of the sphere M and the other end P of the unstretched spring is 25.0 cm, as shown below.



The sphere is rotated at constant speed in a horizontal circle with centre P. The distance PM increases to 26.5 cm.

(a) Explain why the spring increases in length when the sphere is moving in a circle. [2]

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(b) Determine the speed of the sphere. [4]

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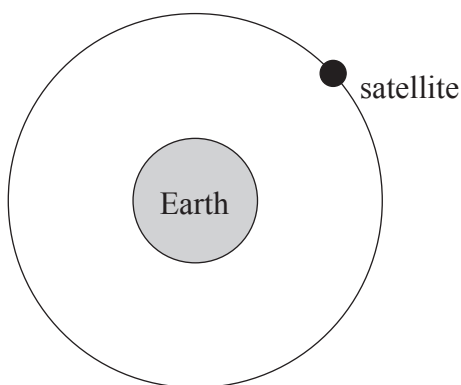
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A3. This question is about circular motion.

A geo-stationary satellite is one that orbits the Earth in an equatorial plane in the same direction of rotation as that of the Earth and with an orbital period of 24 hours. Since the period of rotation of the Earth is 24 hours, this means that the satellite is stationary relative to a point on the Equator.

(a) The diagram below shows a geostationary satellite in orbit about the Earth.

not to scale



On the diagram above, draw an arrow to show the direction of acceleration of the satellite.

[1]

(b) State the name of the force causing the satellite's acceleration.

[1]

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(c) The distance of the satellite from the centre of the Earth is 4.2×10^7 m. Calculate the acceleration of the satellite.

[3]

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A2. This question is about circular motion and global warming.

- (a) A car is travelling at constant speed of 18 m s^{-1} around a horizontal bend in the road. The mass of the car is $1.5 \times 10^3\text{ kg}$ and the bend forms part of a circle of radius $2.0 \times 10^3\text{ m}$.

- (i) State why the car is accelerating.

[1]

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- (ii) Determine the frictional force between the tyres of the car and the surface of the road that produces the acceleration.

[2]

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(Question B3, part 2 continued)

- (e) The force on a mass of 1.0 kg falling freely near the surface of Jupiter is 25 N. The radius of Jupiter is 7.0×10^7 m.

- (i) State the value of the magnitude of the gravitational field strength at the surface of Jupiter. [1]

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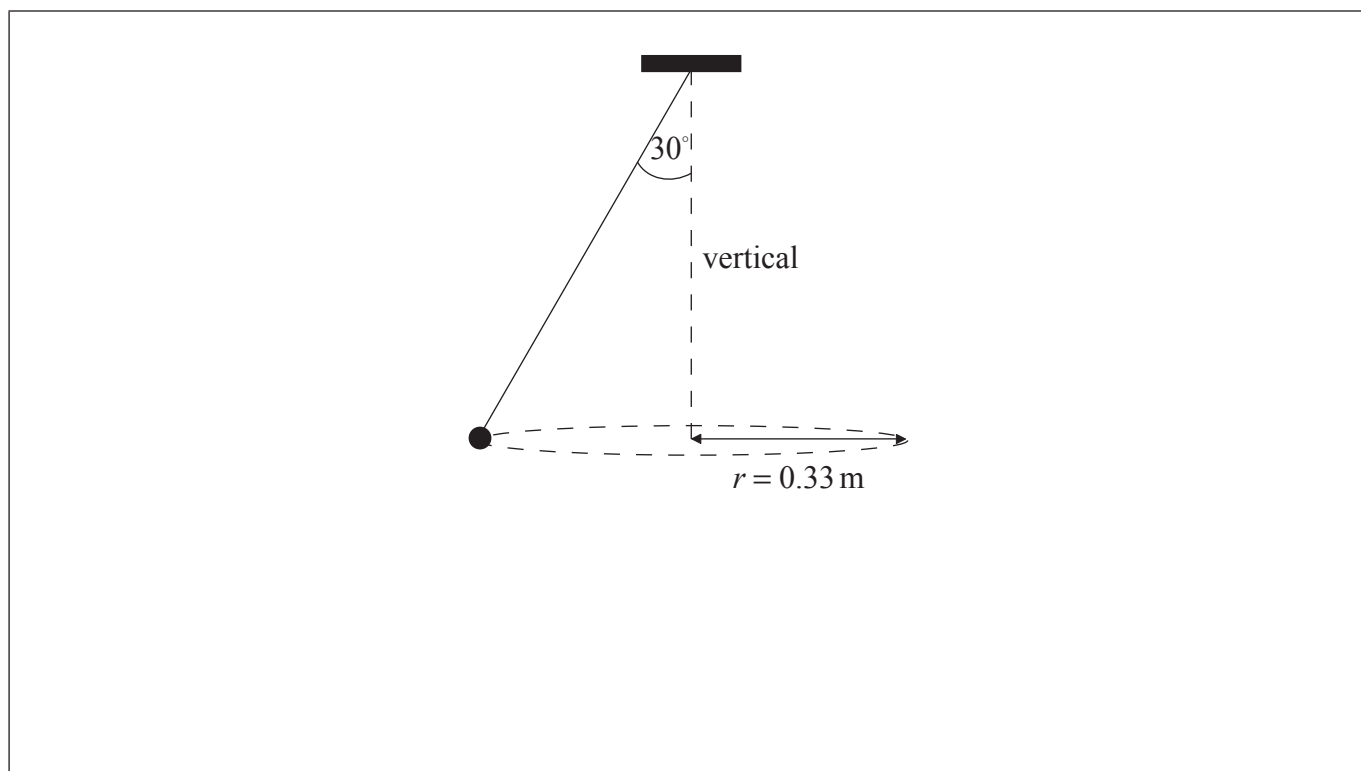
- (ii) Calculate that the mass of Jupiter is about 1.8×10^{27} kg. [2]

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A3. This question is about circular motion.

A ball of mass 0.25 kg is attached to a string and is made to rotate with constant speed v along a horizontal circle of radius $r = 0.33\text{ m}$. The string is attached to the ceiling and makes an angle of 30° with the vertical.



- (a) (i) On the diagram above, draw and label arrows to represent the forces on the ball in the position shown. [2]
- (ii) State and explain whether the ball is in equilibrium. [2]

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(This question continues on the following page)



(Question A3 continued)

(b) Determine the speed of rotation of the ball.

[3]

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(Question B2 continued)

Part 2 Gravitational fields

- (a) State Newton’s universal law of gravitation.

[3]

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- (b) Deduce that the gravitational field strength g at the surface of a spherical planet of uniform density is given by

$$g = \frac{GM}{R^2}$$

where M is the mass of the planet, R is its radius and G is the gravitational constant. You can assume that spherical objects of uniform density act as point masses.

[2]

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(Question B2, part 2 continued)

- (c) The gravitational field strength at the surface of Mars g_M is related to the gravitational field strength at the surface of the Earth g_E by

$$g_M = 0.38 \times g_E.$$

The radius of Mars R_M is related to the radius of the Earth R_E by

$$R_M = 0.53 \times R_E.$$

Determine the mass of Mars M_M in terms of the mass of the Earth M_E .

[2]

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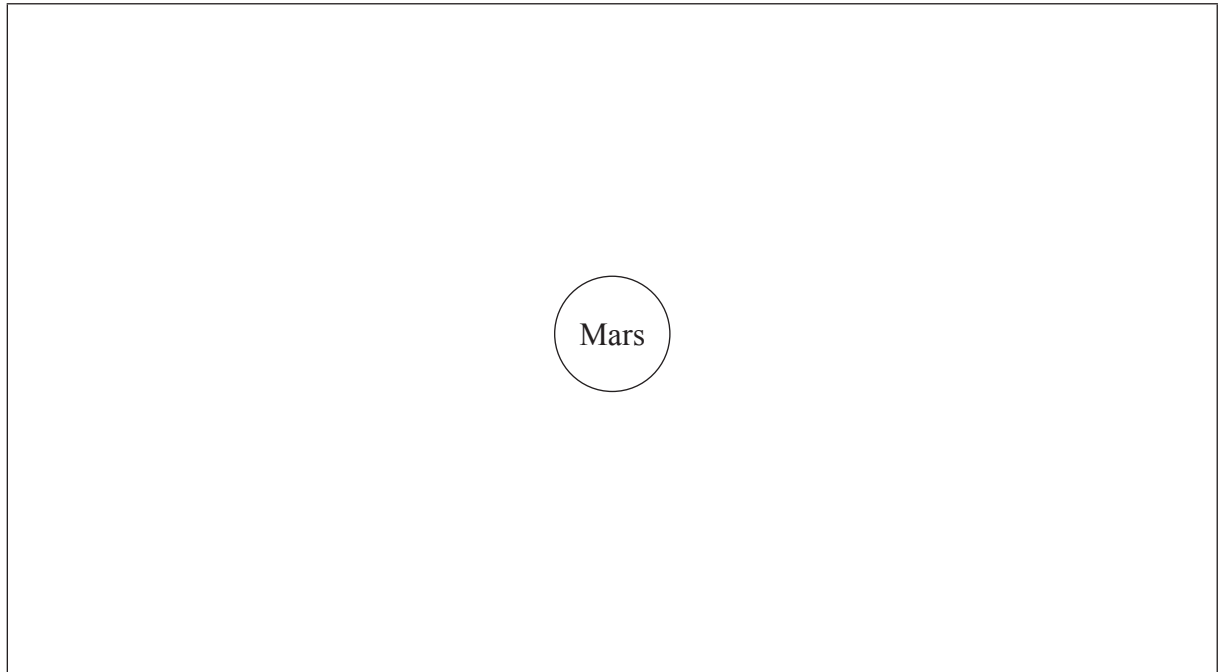
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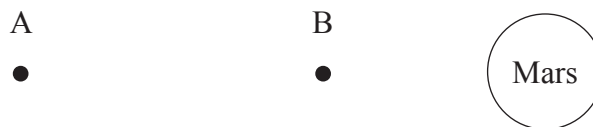


(Question B2, part 2 continued)

- (d) (i) On the diagram below, draw lines to represent the gravitational field around the planet Mars. [1]



- (ii) An object falls freely in a straight line from point A to point B in time t . The speed of the object at A is u and the speed at B is v . A student suggests using the equation $v = u + g_M t$ to calculate v . Suggest **two** reasons why it is not appropriate to use this equation. [2]



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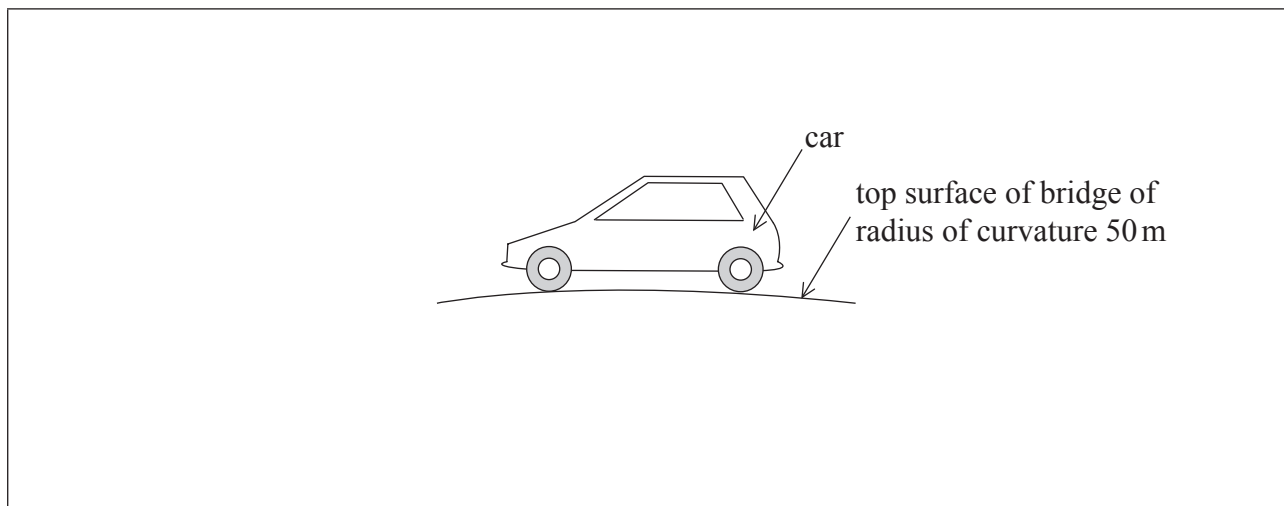
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A2. This question is about circular motion.

The diagram shows a car moving at a constant speed over a curved bridge. At the position shown, the top surface of the bridge has a radius of curvature of 50 m.



- (a) Explain why the car is accelerating even though it is moving with a constant speed. [2]

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- (b) On the diagram, draw and label the vertical forces acting on the car in the position shown. [2]

- (c) Calculate the maximum speed at which the car will stay in contact with the bridge. [3]

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